AVIM PARASARA

Windows Application Developer | C# / WinForms / WPF | Industrial HMI & CNC Control | Embedded Systems & OpenGL
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PROFESSIONAL SUMMARY

Windows Application Developer with hands-on experience building industrial-grade desktop software and embedded communication systems in C# and WinForms. Built Proton HMI, a generalized CNC interface unifying Ethernet communication across FANUC, Siemens, and Syntec controllers, and separately developed a GPU-accelerated OpenGL digital twin and G-code simulator. Experienced across the full stack of a CNC control system - from low-level sensor and microcontroller communication (ESP32, Arduino, STM32, RS-485) to real-time HMI interfaces and 3D visualisation.

WORK EXPERIENCE

Junior Software Developer - Enliven Digital Content *Jul 2025 - Present*

- Build Proton HMI, a generalized industrial CNC interface (C#/WinForms) that unifies Ethernet communication across FANUC, Siemens, and Syntec controllers, alongside a GPU-accelerated OpenGL digital twin and G-code simulator for real-time 3D visualisation.
- Design a multi-threaded data pipeline (read, parse, model, then dispatch to the UI) shared across all three controller integrations, and resolve intermittent ~1-second latency spikes on safety-critical signal reads.
- Own architecture decisions around threading, buffering, and controller-agnostic module design, plus rendering-correctness fixes and OpenGL-to-C# P/Invoke interop debugging.

Windows Developer Trainee - Enliven Digital Content *Jun 2025 - Jul 2025*

- Onboarded onto the CNC/HMI project stack, learning C# desktop development, embedded communication fundamentals, and the existing WinForms architecture under senior guidance.
- Tested sensor and microcontroller communication in the early build phase before taking on independent feature ownership.

.NET Application Developer (Internship) - Myalgomate Technologies LLP *Mar 2024 - Jun 2024*

- Built algorithmic trading strategy execution software in C# and WinForms enabling automated order placement and real-time position management.
- Integrated APIs of multiple stockbrokers for live order routing, market data feeds, and position tracking across different broker formats.
- Implemented response parsing and data normalisation logic to create a unified internal data model across all broker integrations.
- Maintained production-ready codebase through clean coding practices, structured debugging, and code reviews.

KEY PROJECTS

Proton HMI - Multi-Controller CNC Interface - *C#, WinForms, Ethernet, Multi-threading, FANUC / Siemens / Syntec*

- Built a generalized WinForms CNC interface that standardises Ethernet communication across FANUC, Siemens, and Syntec controllers, streaming axis/work/machine coordinates, feed and spindle overrides, PMC signals, tool data, alarms, and program state in near real-time.
- Engineered a multi-threaded polling pipeline (read, parse, model, then dispatch to the UI) shared across all three integrations, and diagnosed intermittent ~1-second latency spikes on safety-critical signal reads.

Digital Twin & G-Code Simulator - *C#, OpenGL (P/Invoke), Shaders, 3D Mesh Rendering - GPU-Accelerated Visualisation*

- Developed a digital twin of the CNC machine with a G-code simulator (G01 linear and G02/G03 arc toolpaths, stock/tool rendering, live tool animation, ViewCube navigation); resolved Z-fighting, draw-order, and depth-buffer defects, and debugged C#-to-OpenGL P/Invoke interop issues.

Industrial Sensor Communication Pipeline - *Node.js, Revolution Pi (headless), ESP32 (x2), Arduino, MAX485 / RS-485, SQLite*

- Built a position-sensor and temperature-monitoring pipeline in Node.js running headless on a Revolution Pi, interfacing two ESP32 modules, an Arduino, and a MAX485 RS-485 transceiver - collecting live data from 25 DS18B20 temperature sensors and 16 M18 inductive proximity (position) sensors every 600 ms into SQLite for continuous industrial monitoring feeding the HMI.

DIY CNC Machine - Custom HMI Integration - *STM32, NEMA17 & NEMA23 Stepper Motors, Ethernet, C#, WinForms*

- Built a DIY CNC machine (STM32 + NEMA17/NEMA23 stepper motors) driven by a custom HMI over Ethernet, achieving ~5 ms round-trip response for smooth, low-latency motor control.

TECHNICAL SKILLS

Languages & DB: C#, C (embedded), MS SQL Server, SQLite
Desktop UI: WinForms, WPF, Custom Controls, MVVM
Graphics: OpenGL via P/Invoke, GPU-accelerated rendering, shaders, 3D mesh rendering
Industrial Protocols & Hardware: FANUC/Siemens/Syntec Ethernet, RS-485/MAX485, PMC signals, ESP32, Arduino, STM32, Revolution Pi, NEMA17/23 Steppers, DS18B20, M18 Inductive Proximity
Speciality: HMI Development, CNC Control, Real-Time Systems, Digital Twin Visualisation
Tools & Practices: Visual Studio, Git, OOP, Multi-threading

ACHIEVEMENTS

Employee of the Quarter - Enliven Digital Content | Jan-Mar 2026 - recognised for valuable insight and contributions during the quarter.

EDUCATION

Bachelor of Computer Applications - Geetanjali College | 2021 - 2024 | CGPA 7.56
H.S.C - Gujarat State Board | 2021 | 68% **S.S.C** - Gujarat State Board | 2019 | 69% **Languages:** Gujarati, Hindi, English (Fluent)